



Name: Lavlin Jaison

Business Function: GenAI

Project Title:

Building a Knowledge Graph-Powered Alert & Chatbot System for Macro-Pricing Intelligence using Neo4j and LLM

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Buddy: Arnab Chaudhary



NextGen Internship Program FY 26-27

OBJECTIVES



- Read Datatables
- Filter redundancy
- Standardise values

DATA CLEANING



- Define Nodes
- Define Relationships
- Setup Connection
- Fill in filtered data

GRAPH BUILDING



- Currency Alerts
- Dynamic Theshold
- Commodity Alerts
- Query Relevant AIs

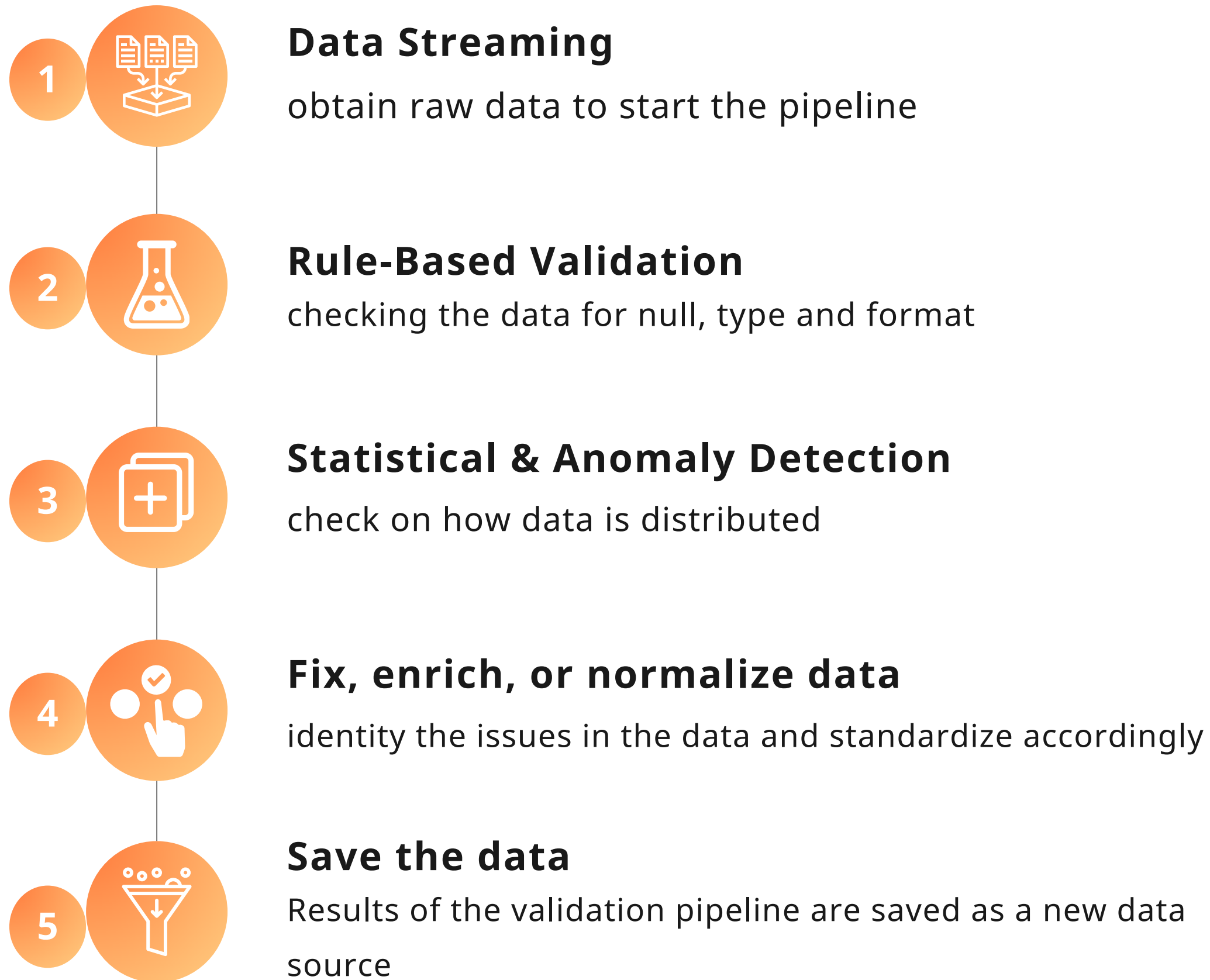
ALERT SYSTEM



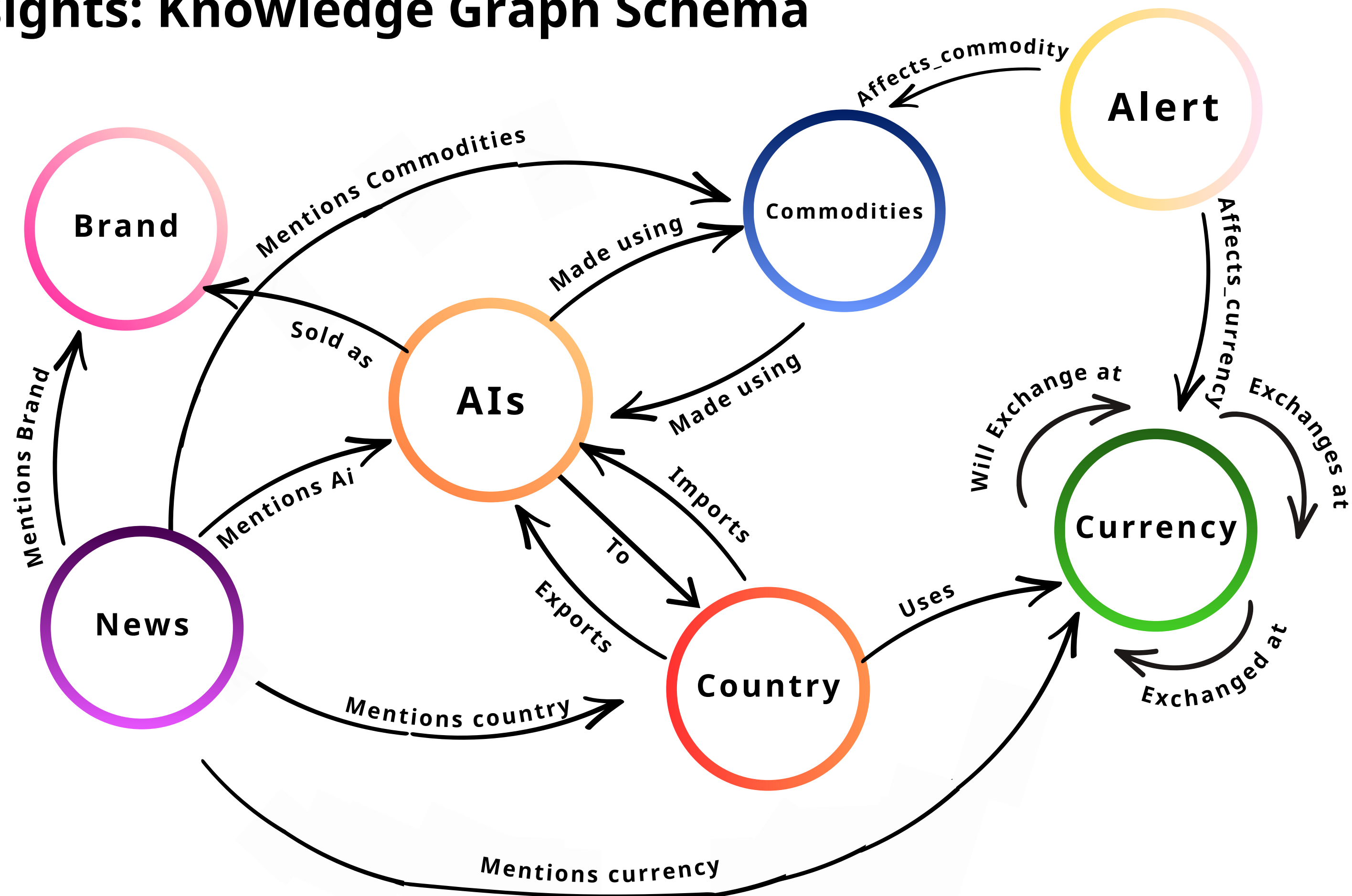
- Multi stage Architecture
- Connection identification
- Visualisation Tool

GRAPH RAG SYSTEM

DATA VALIDATION PIPELINE



Insights: Knowledge Graph Schema



Graph Properties

- Node Properties

Node Label	Property Name	Data Type	Description	Datatable
Commodity	name	String	Full name of the commodity (e.g., Soybean, Ammonia)	commodity_future_forecast
	symbol	String	Shorthand market ticker/code	commodity_future_forecast
	unit	String	Unit of measurement used (e.g., Metric Tons, Bushels)	commodity_future_forecast
	Price	Float	Latest recorded spot price	commodity_future_forecast
	forecast_price	List of Floats	Forecasted prices for the next four financial quarters	commodity_future_forecast
	forecast_date	List of Strings	Target dates/quarters for forecasted pricing	commodity_future_forecast



Graph Properties

- Node Properties

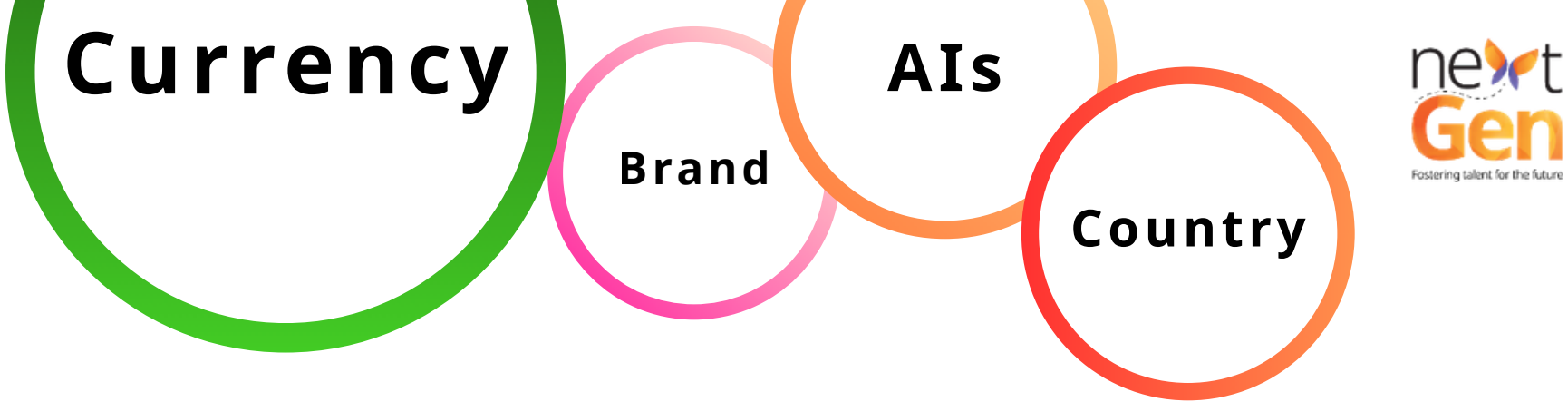
Node Label	Property Name	Data Type	Description	Datatable
News	article_number	Float	Number given to the news article (e.g. '0')	agro_news_ext
	Upl_Relevant	String	Whether news is relevant to Upl or not (e.g. "Yes" or "No")	agro_news_ext
	Title	String	Name given to the news article (e.g. "Get Ahead of China's Next Industrial Shift -Before Your Competitors Do")	agro_news_ext

Graph Properties

- Node Properties



Node Label	Property Name	Data Type	Description	Datatable
Alerts	Severity	String	How severely it affects Commodities and Currency (e.g. "CRITICAL")	
	Created_at	String	When was the alert created (e.g. "2026-07-22")	
	Id	String	Unique Identifier generated to identify alert (e.g."f6237eef-48fd-4519-b4fd-91bc7d8d7ed9")	
	Title	String	Name of alert (e.g."Extreme price spike of 7.94% to 4639.20 CNY/T by 2026-09-30")	
	Alert_type	String	Type of Alert (e.g. "EXTREME SURGE")	
	Status	String	Current status of alert (e.g. "active")	



• Node Properties

Node Label	Property Name	Data Type	Description	Datatable source
Currency	name	String	Three-letter ISO currency code (e.g., USD, BRL, INR)	currency_exchange_rates_v2, currency_future_forecast
Country	name	String	Standard name of the geopolitical entity	currency_exchange_rates_v2, currency_future_forecast
Active_Ingredient	name	String	Chemical name of the active molecule (AI)	monthly_ais
Brand	name	String	Commercial brand name associated with active ingredients	material_master

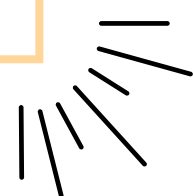


- Relationship Properties

Relationship Name	From Node	To Node	Properties Attached
EXCHANGES_TO	Currency	Currency	date (str), exchange_rate (float)
EXCHANGED_TO	Currency	Currency	dates (list of str), rates (list of float)
WILL_EXCHANGE_TO	Currency	Currency	dates (list of str), rates (list of float) [FX Forecasts]
IMPORT	Country	Active_Ingredient	dates (list of str), weight_kg (list of float), Amount_dollars (list of float), id (float)
EXPORT	Country	Active_Ingredient	dates (list of str), weight_kg (list of float), Amount_dollars (list of float), id (float)

• Relationship Properties

Relationship Name	From Node	To Node	Description
TO	Active_Ingredient	Country	id (float) [Identifies the endpoint country]
USES	Country	Currency	No properties [Denotes national currency standard]
SOLD_AS	Active_Ingredient	Brand	material_cd (list of str) [Unique identifiers of physical products]
MADE_USING	Active_Ingredient	Commodity	reason (str) [Synthesized LLM]
MADE_USING	Commodity	Active_Ingredient	reason (str) [Synthesized LLM]



- Relationship Properties

Relationship Name	From Node	To Node	Description
Mentions Currency	News	Currency	No properties attached
Mentions Country	News	Country	No properties attached
Mentions Brand	News	Brand	No properties attached
Mentions Ai's	News	Active_Ingredient	No properties attached
Mentions Commodities	News	Commodities	No properties attached

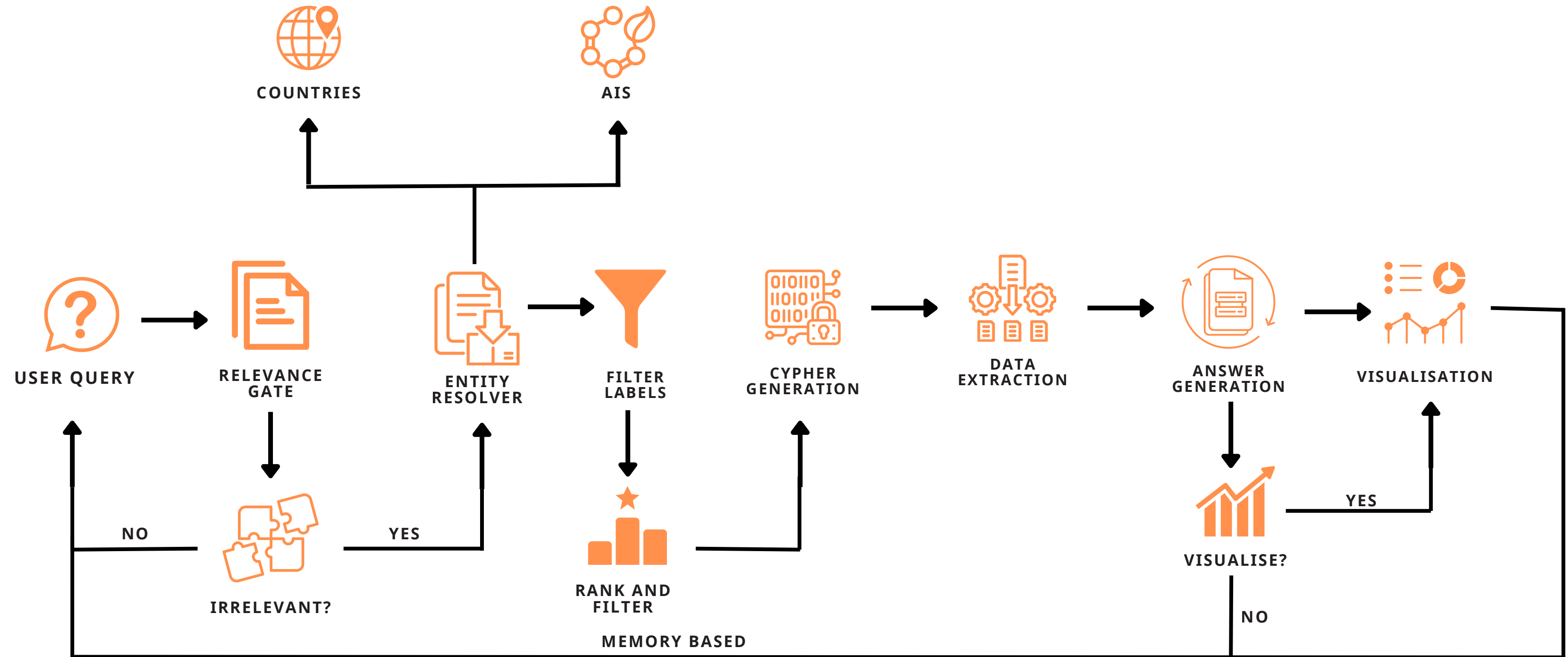


- Relationship Properties

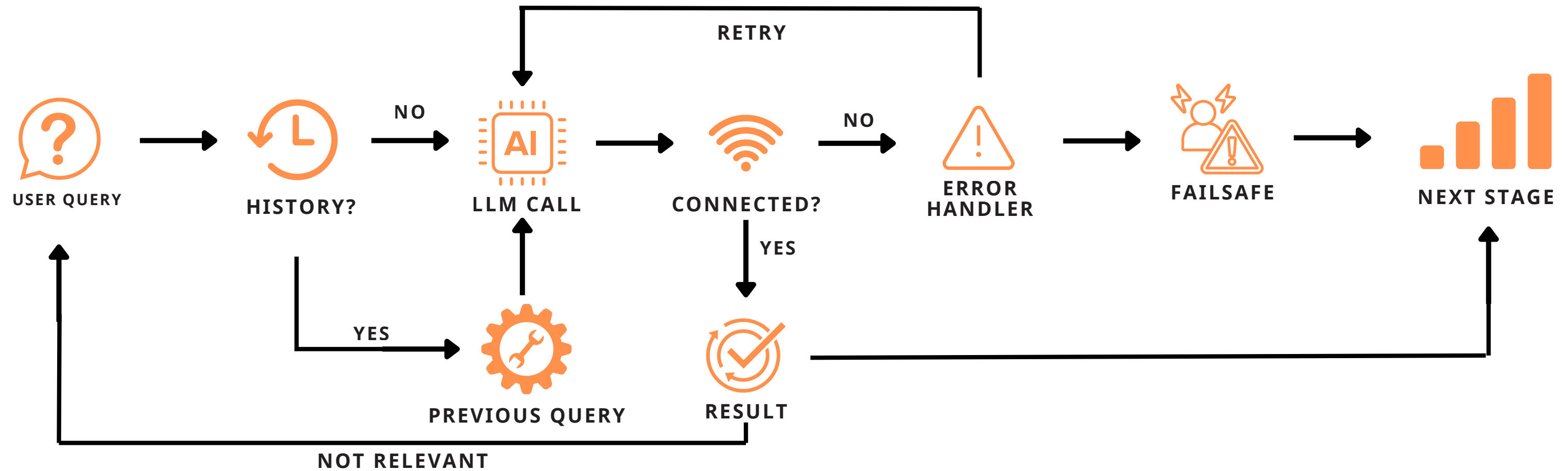
Relationship Name	From Node	To Node	Description
Affects Currency	Alerts	Currency	No properties attached
Affects Commodities	Alerts	Commodities	No properties attached



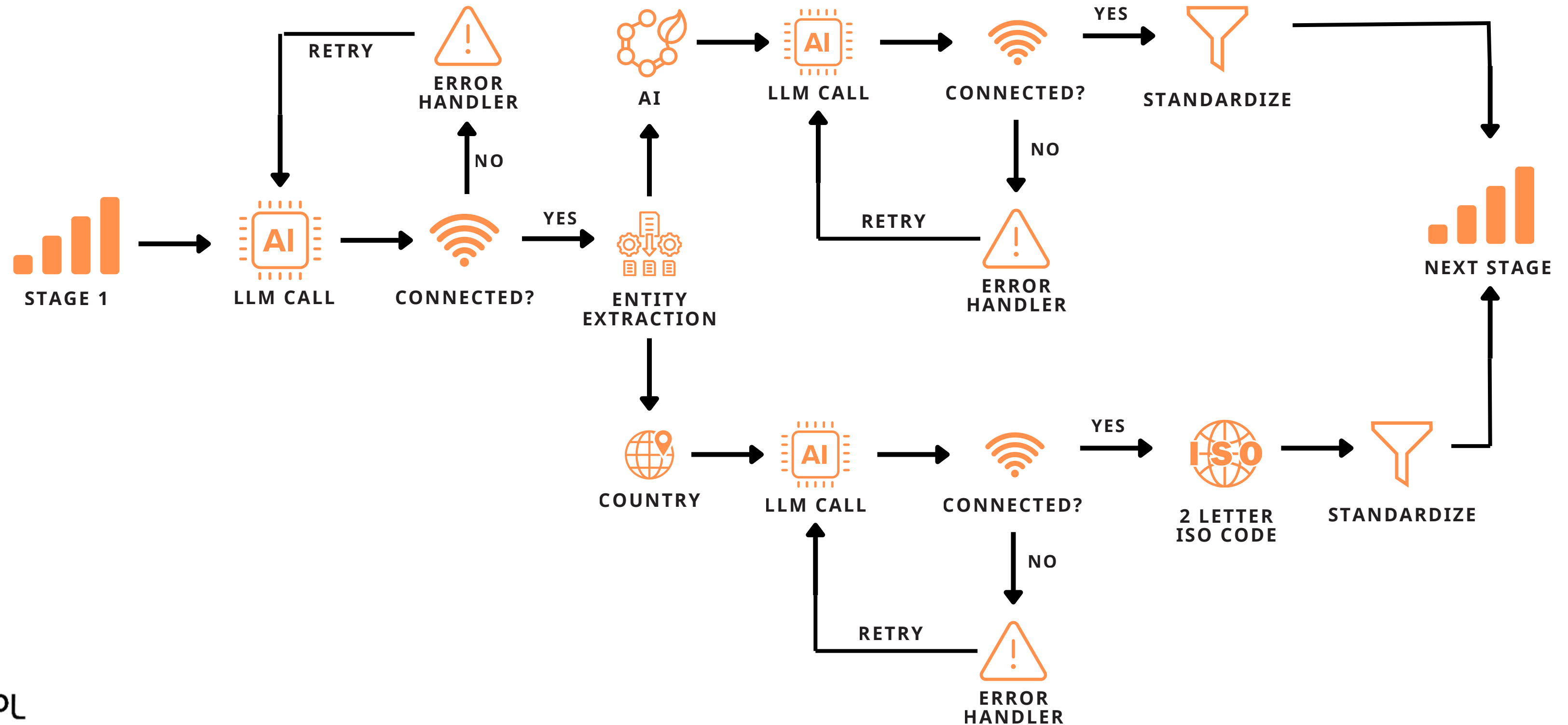
RAG PROGRAM FLOW



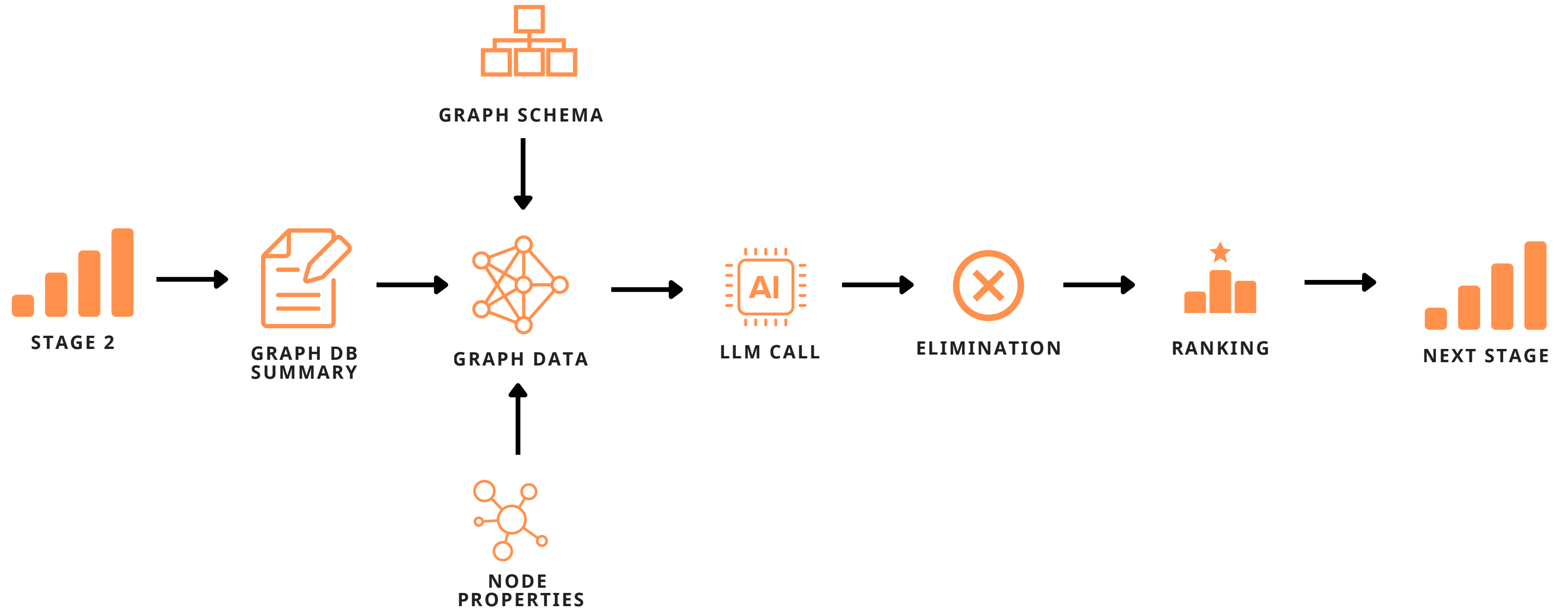
1. RELEVANCE GATE



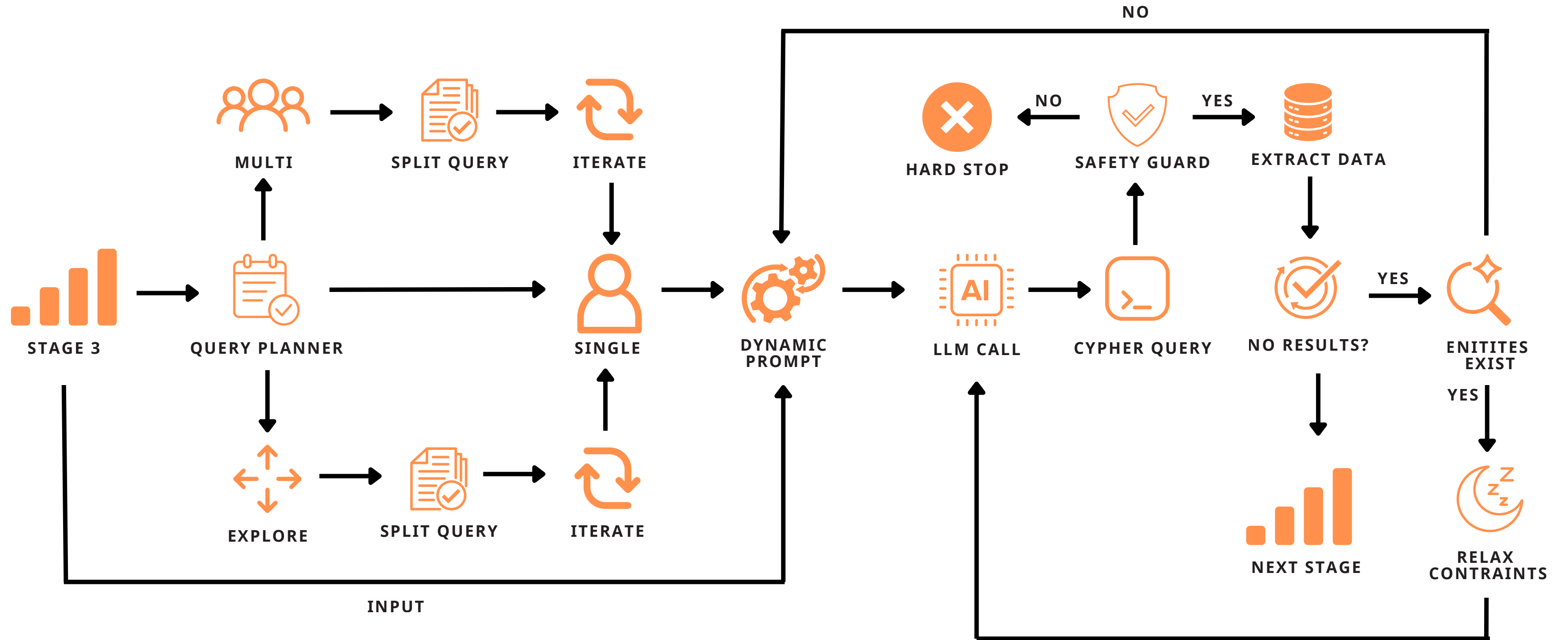
2.ENTITY RESOLVER



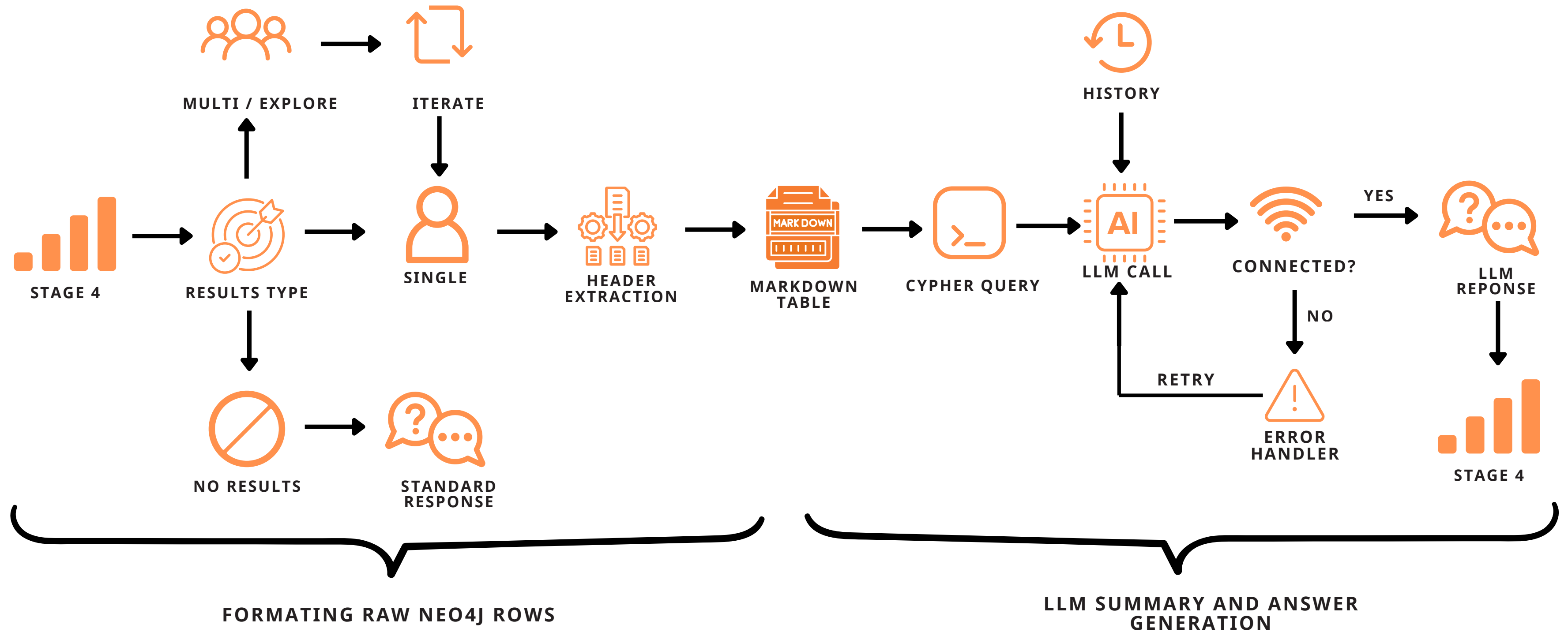
3. FILTER LABELS



4.CYPHER GENERATION



5.ANSWER GENERATION



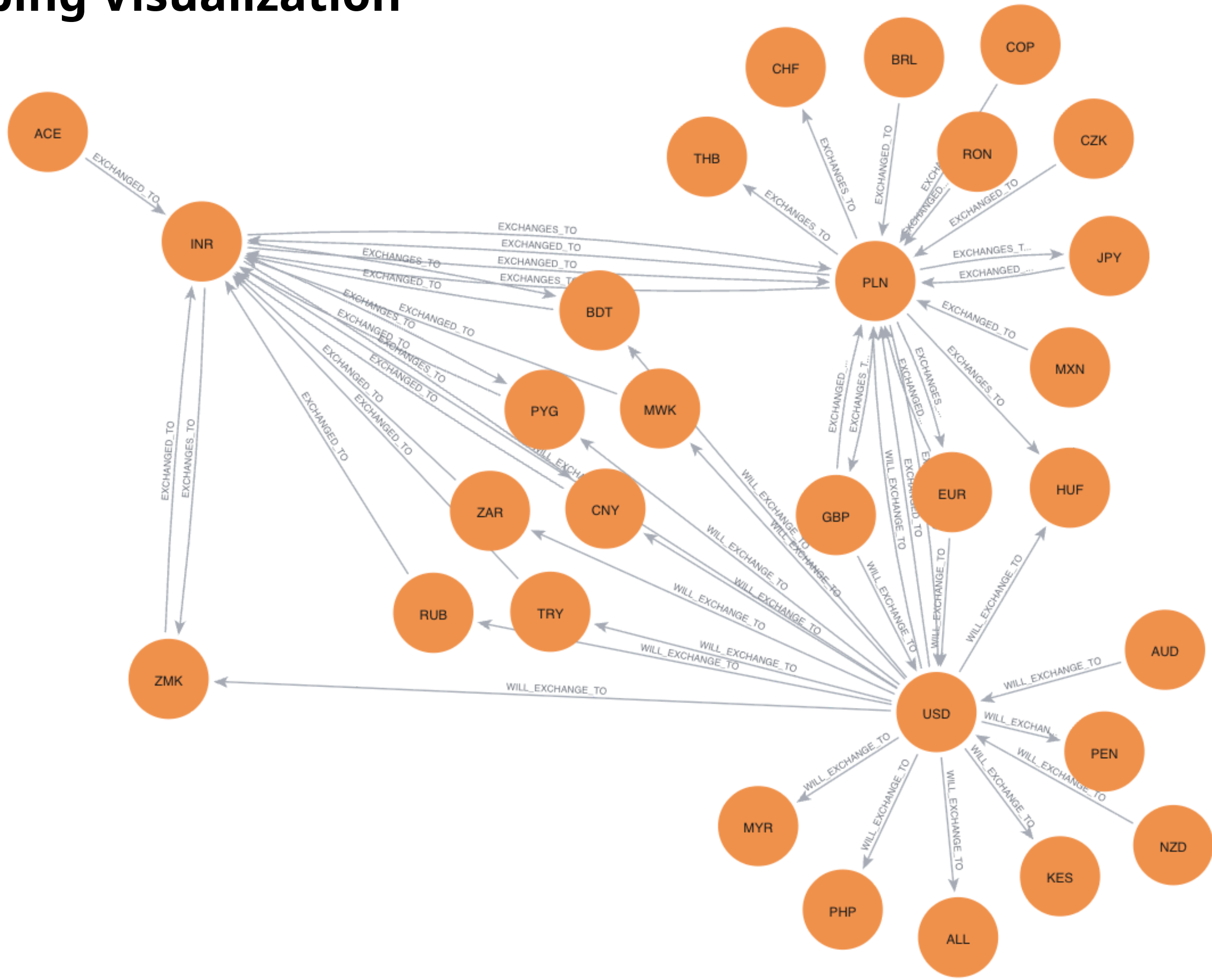


RECOMMENDATIONS

- 1 Direct Access to databricks using Spark Connector
- 2 Access to neo4j Aura for hosting graph database in cloud
- 3 Automatic updating of graph database
- 4 Adding more data into graph database using more data tables
- 5 Implementing more Alert systems based on graph database
- 6 Additional Testing and finetuning of Graph RAG
- 7 Feasibility with exsisting deep insights architecture

QUESTIONS AND QUERIES

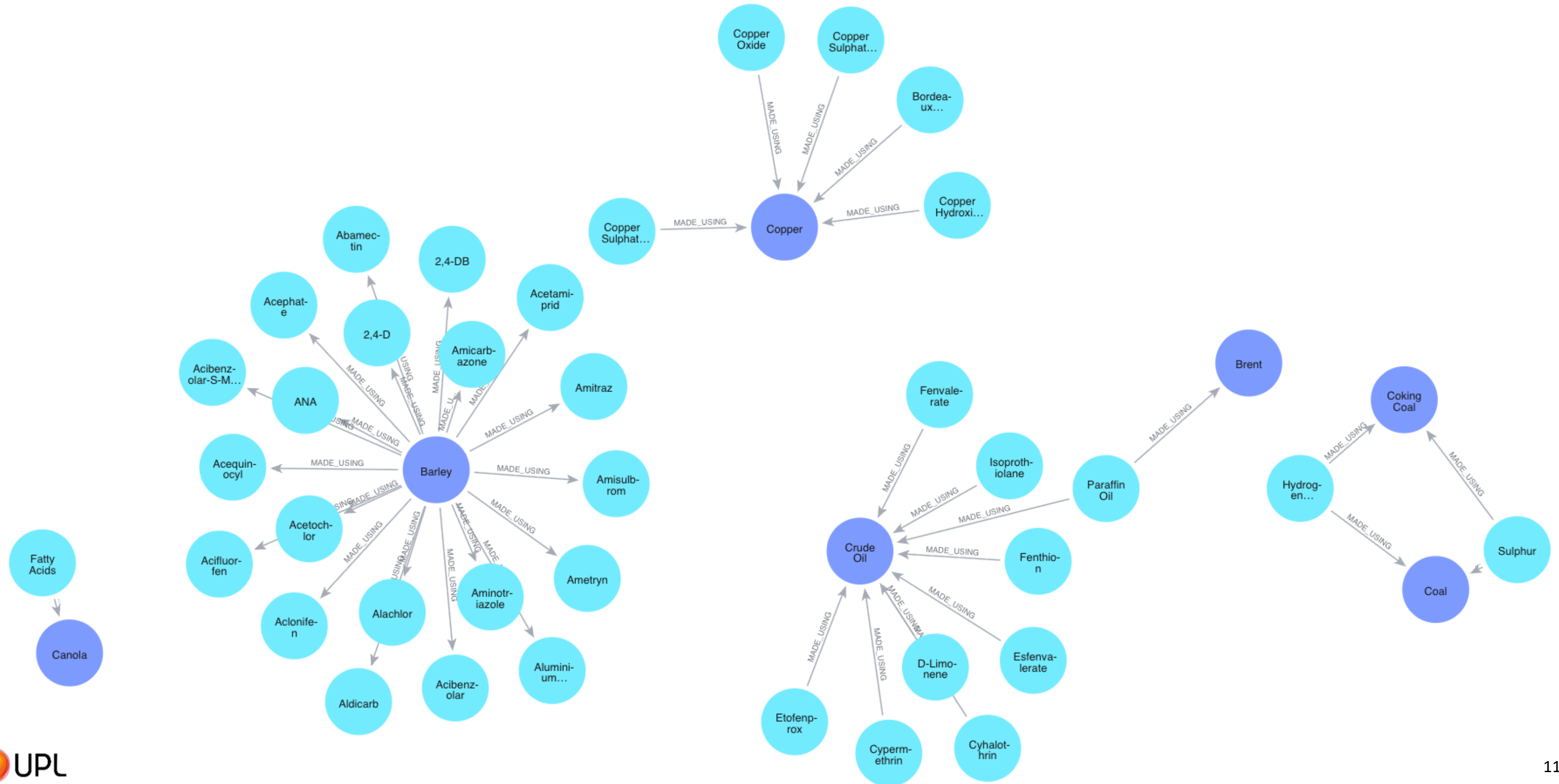
Currency Mapping Visualization



AIs Import and Export Mapping Visualization

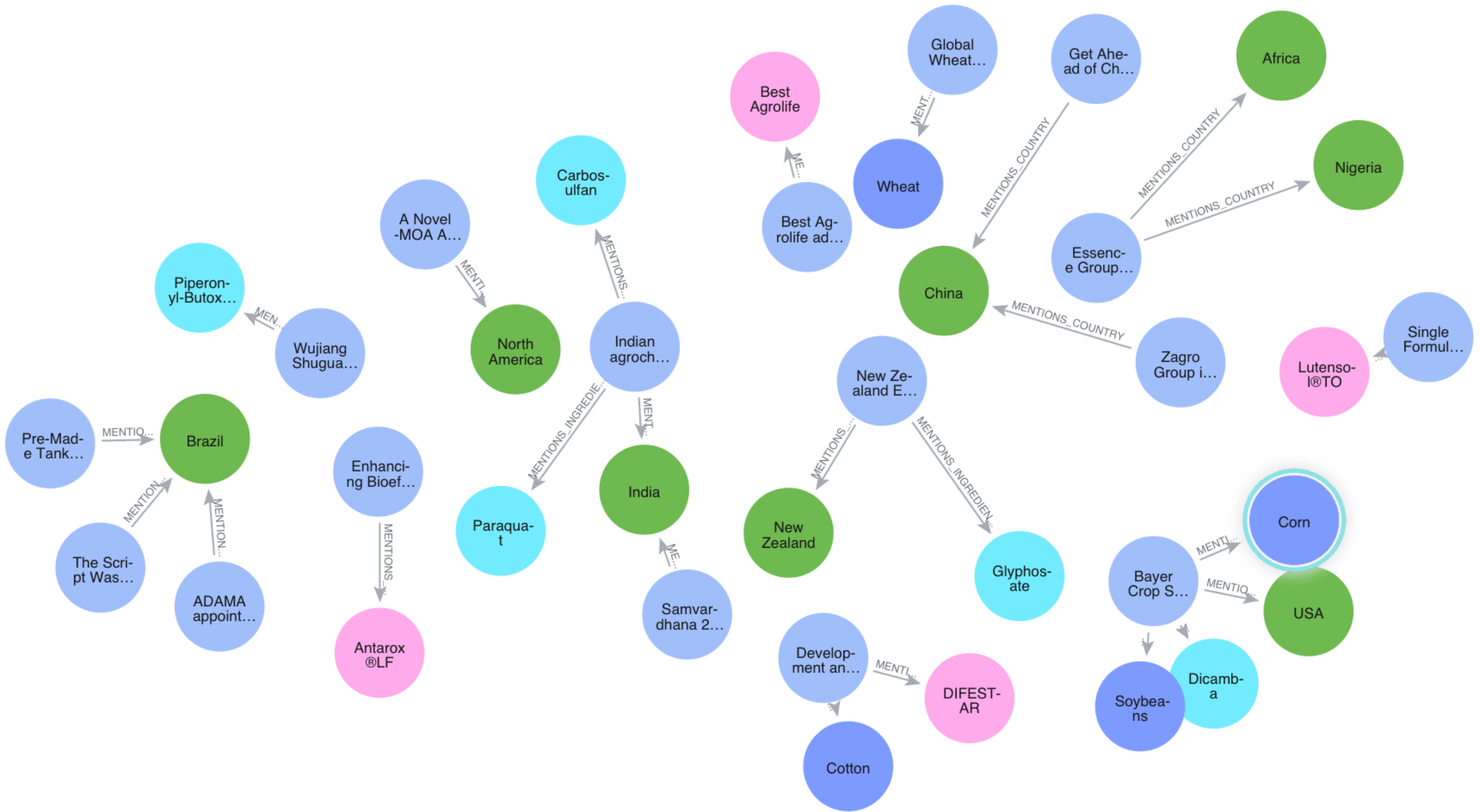


AIs to Commodity Mapping Visualization

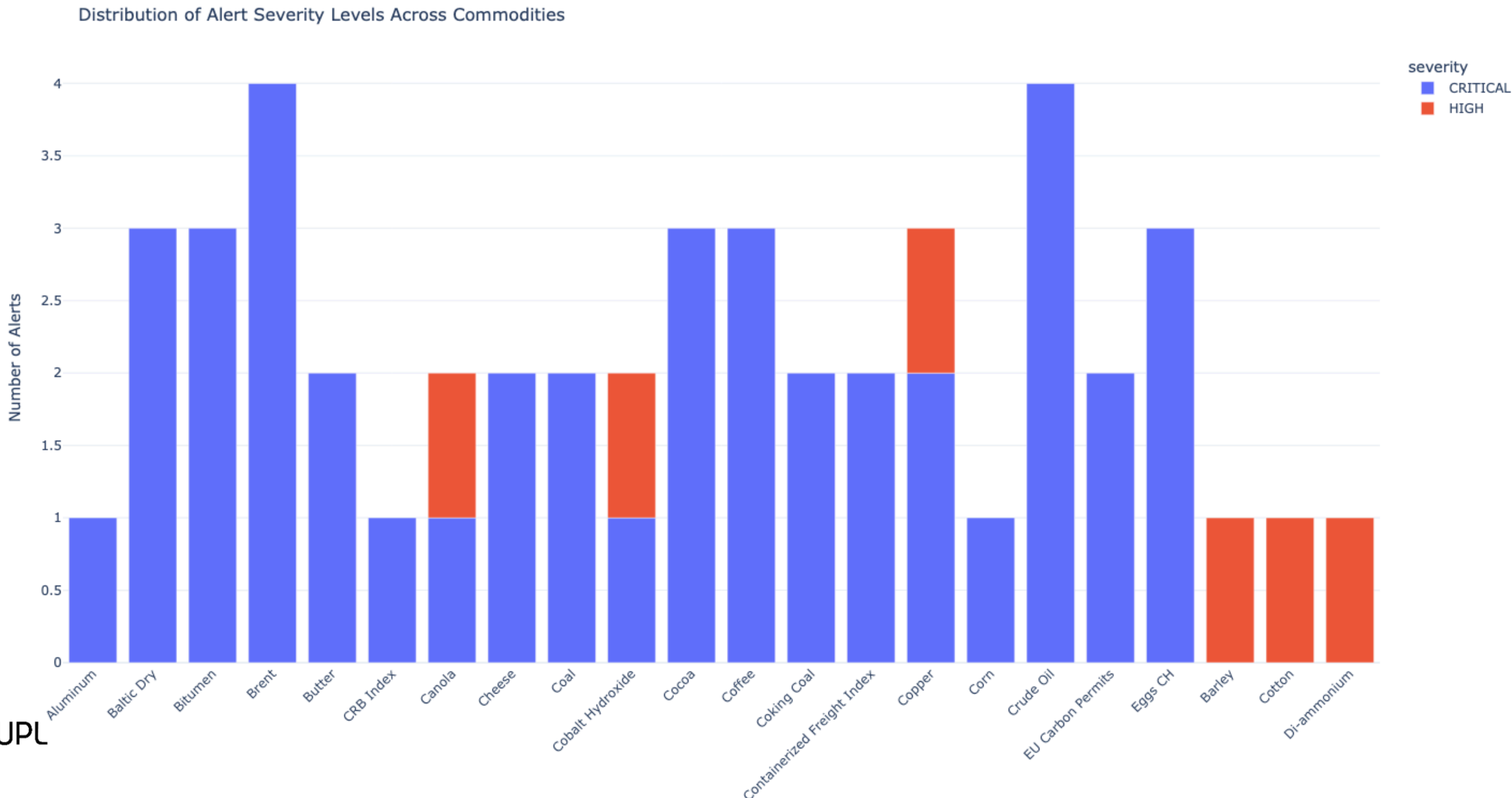


Alert to Commodity Mapping Visualization

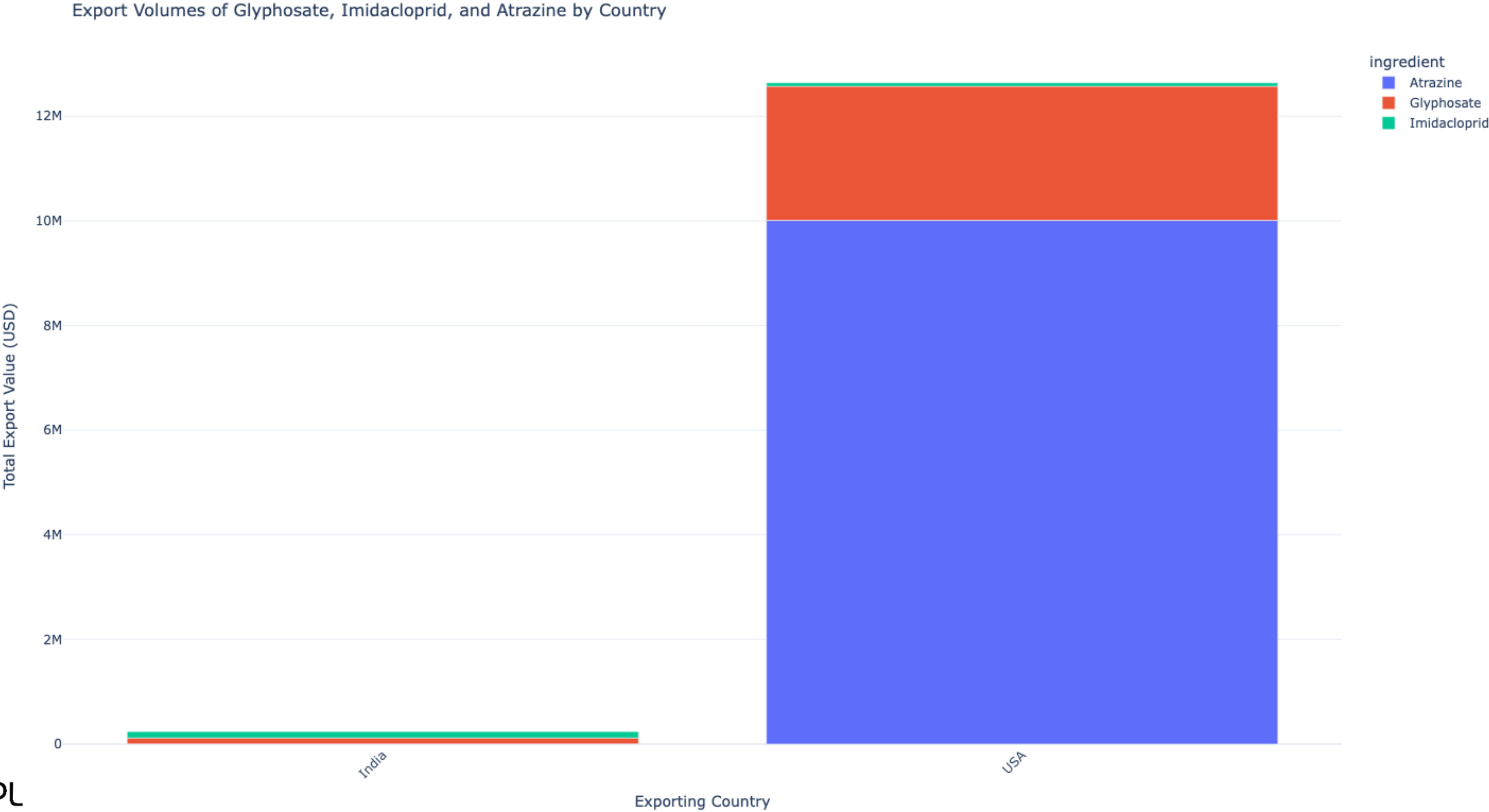




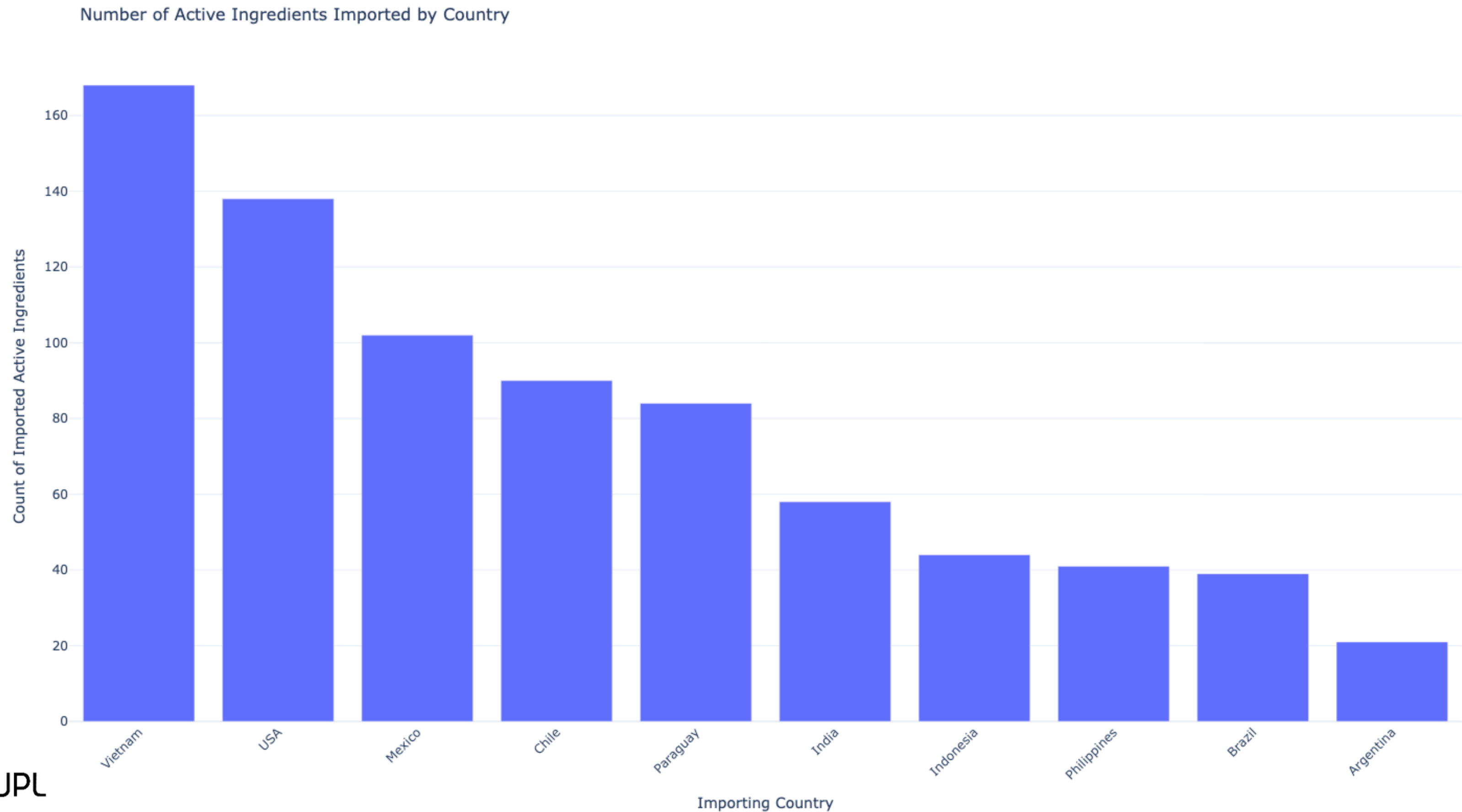
RAG Visualization



RAG Visualization



RAG Visualization



THANK YOU!